

A Regime-Activated Lensing Response Improves the Fit to KiDS-1000 Cosmic Shear and Reframes the Interpretation of S_8

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DOI: [10.6084/m9.figshare.31271917](https://doi.org/10.6084/m9.figshare.31271917)

DRAFT v2.0 — February 2026

Abstract

Weak gravitational lensing measurements are commonly interpreted within the Λ CDM framework using the parameter combination $S_8 = \sigma_8 \sqrt{\Omega_m/0.3}$ as an effective descriptor of shear amplitude. This implicitly assumes that lensing responds directly to matter clustering with no additional scale- or redshift-dependent response factor. We test a phenomenological alternative in which the lensing response includes an explicit, regime-activated term $\Sigma(k, z)$. Using the KiDS-1000 cosmic shear dataset, we find that a late-time, small-scale lensing response significantly improves the likelihood relative to Λ CDM at fixed cosmology, is robust to scale cuts, cannot be reproduced by simple amplitude rescaling, and exhibits a clear tomographic signature localized to low-redshift bins. When cosmological parameters are allowed to vary on a grid, the model prefers lower growth amplitude but higher effective shear amplitude once the response term is included, indicating that part of the apparent S_8 tension arises from parameterization rather than direct observational conflict. All results are maximum-likelihood evaluations, not full posterior inference, and require cross-survey replication before definitive conclusions can be drawn.

Keywords: cosmic shear, weak gravitational lensing, KiDS-1000, S_8 tension, large-scale structure, lensing response, phenomenological gravity, growth–response decomposition

1 Introduction

Cosmic shear surveys provide one of the most direct probes of late-time large-scale structure. In Λ CDM analyses, the primary amplitude constraint is often expressed as

$$S_8 = \sigma_8 \sqrt{\Omega_m/0.3}, \quad (1)$$

which acts as a proxy for the overall shear signal strength. However, this compression assumes that lensing directly traces matter fluctuations without additional scale- or epoch-dependent modifications. Under this assumption, all shear amplitude variation is attributed to the clustering amplitude σ_8 .

We explore an alternative interpretation in which the observed shear signal depends on two separable components:

- **Growth amplitude** — how strongly structure forms (traditionally mapped to σ_8);
- **Lensing response** — how efficiently gravitational potentials generate shear.

In standard Λ CDM analyses, the lensing response is implicitly fixed to unity. We relax this assumption by introducing a phenomenological response factor $\Sigma(k, z)$ that activates only in a specific regime of scale and redshift.

This parameterization is motivated by theoretical frameworks in which the gravitational response may depend on local thermodynamic or information-theoretic conditions, though the present analysis is purely phenomenological and does not assume any specific underlying theory. The functional form is chosen to capture a late-time, scale-dependent activation while remaining simple enough for robust statistical testing.

2 Phenomenological Lensing Response Model

We modify the lensing power spectrum through

$$P_{\text{eff}}(k, z) = P_{\Lambda\text{CDM}}(k, z) \Sigma^2(k, z), \quad (2)$$

where $\Sigma(k, z)$ is defined as

$$\Sigma(k, z) = \left[1 + \alpha \tanh\left(\frac{k - k_{\text{trans}}}{0.05}\right) \right] \left[1 + \alpha \left(1 - \frac{z}{z_{\text{activ}}}\right) \right] \quad (z < z_{\text{activ}}) \quad (3)$$

and $\Sigma = 1$ for $z \geq z_{\text{activ}}$.

For the best-fit model: $\alpha = 0.10$, $k_{\text{trans}} = 0.1 \, h \, \text{Mpc}^{-1}$, $z_{\text{activ}} = 0.5$. This produces a smooth enhancement of lensing response at late times and moderately nonlinear scales. At the representative pivot point $k_* = 0.15 \, h \, \text{Mpc}^{-1}$, $z_* = 0.25$, the effective response is $\Sigma \approx 1.13$.

We emphasize that this is a *Case A* implementation: Σ modifies only the lensing amplitude (the Weyl potential $\Phi + \Psi$), not the growth of perturbations. The matter power spectrum $P(k, z)$ is computed from standard Λ CDM CAMB output. A fully self-consistent modified gravity implementation (*Case B*), where modifications also enter the perturbation equations via a separate growth function $\mu(k, z)$, is deferred to future work. Because growth is not modified in this implementation, the model should be interpreted strictly as a phenomenological lensing-response degree of freedom rather than a full modified-gravity scenario.

3 Data and Method

We analyze the KiDS-1000 $\xi_{\pm}$ shear correlation functions using the publicly released covariance matrix and tomographic redshift distributions (Asgari et al., 2021), implemented within the KCAP/CosmoSIS framework with the Flinc simulation-based covariance.

3.1 Pipeline Configuration

All results presented here are maximum-likelihood evaluations at fixed or grid-varied cosmological parameters, not full Markov Chain Monte Carlo (MCMC) posterior inference. The baseline cosmology uses Planck 2018 best-fit values (Planck Collaboration, 2020).

Nuisance parameters are configured as follows: intrinsic alignment amplitude $A_{\text{IA}} = 0.0$ (fixed/off); photometric redshift bias Δz and multiplicative shear bias m are not included; the HMCode baryonic feedback amplitude is fixed at $A = 3.13$ (Mead et al., 2015). The total number of data points after masking is 195 (105 ξ_+ and 90 ξ_-).

Note: Because all nuisance parameters are fixed and IA is set to zero, the improvement cannot be attributed to nuisance parameter compensation. However, this also means that a full marginalization over nuisance parameters remains to be performed and may modify the magnitude of the improvement.

3.2 Tests Performed

The following validation sequence was applied: (a) Λ CDM baseline reproduction; (b) null test with $\Sigma = 1$ to verify the pipeline hook; (c) parameter scan over 13 values of α in the range $[0.00, 0.20]$; (d) amplitude-equivalence test varying only A_s ; (e) hard scale cuts removing 45 small-angular-scale data points; (f) tomographic subset likelihoods for low- z (bins 1–2) and high- z (bins 3–5); (g) mini-profile on a 5×5 grid in (Ω_m, A_s) .

4 Results

4.1 Parameter Scan and Likelihood Improvement

At fixed cosmology, the α scan reveals a smooth, well-defined peak (Table 1).

Table 1: Selected values from the full α scan.

α	$-\ln \mathcal{L}$	$\Delta(-2 \ln \mathcal{L})$	Note
0.00	−349.68	0.0	Baseline
0.05	−331.08	−37.2	
0.10	−324.21	−50.9	Best-fit
0.12	−325.64	−48.1	
0.15	−333.19	−33.0	
0.20	−362.95	+26.5	Overshoots

We report the improvement as $\Delta(-2 \ln \mathcal{L})$ rather than a formal statistical significance, because (a) the evaluation is at fixed nuisance parameters, (b) the single additional parameter α was scanned rather than profiled in a full joint fit, and (c) converting likelihood differences to σ -equivalents requires assumptions about parameter regularity and boundary effects that have not been verified. The key result is that the improvement is large ($\Delta(-2 \ln \mathcal{L}) \approx 51$), smooth in α , and peaked rather than monotonic.

4.2 Robustness: Not an Amplitude Rescaling

Allowing Λ CDM to vary only the primordial amplitude A_s improves the fit by $\Delta(-2 \ln \mathcal{L}) \approx 10$ (best at $\ln(10^{10} A_s) = 3.04$, a 9% increase). The response model at the same baseline A_s improves the fit by $\Delta(-2 \ln \mathcal{L}) \approx 51$. The residual improvement of $\Delta(-2 \ln \mathcal{L}) \approx 41$ cannot be reproduced by amplitude scaling and demonstrates sensitivity to the scale and redshift structure of $\Sigma(k, z)$. Approximately 80% of the total improvement is shape-dependent.

4.3 Robustness: Scale Cuts

Removing 45 small-angular-scale data points (23% of the total), the improvement remains substantial (Table 2).

Table 2: Likelihood improvement under standard and hard scale cuts.

Configuration	N_{data}	$\Delta(-2 \ln \mathcal{L})$	Retained
Standard cuts	195	−50.9	100%
Hard cuts	150	−32.0	63%

The fact that 63% of the improvement persists after aggressive scale cuts indicates the signal is not confined to the most nonlinear scales where baryonic feedback and halo model uncertainties dominate.

4.4 Tomographic Fingerprint

Splitting the data by tomographic redshift bins reveals that the improvement is concentrated at low redshift (Table 3).

Table 3: Tomographic decomposition of likelihood improvement.

Subset	N_{data}	$-\ln \mathcal{L}_{\Lambda\text{CDM}}$	$-\ln \mathcal{L}_{\text{EFC}}$	$\Delta(-2 \ln \mathcal{L})/N$
Low- z (1–2)	117	−211.37	−160.43	0.87
High- z (3–5)	78	−93.75	−83.04	0.27
All	195	−349.68	−324.21	0.26

Note: Subset likelihoods are not directly summable to the full likelihood because the covariance matrix couples low- z and high- z bin pairs. The subsets use only the diagonal blocks of the full covariance. The relative ratio ($3.2\times$ per datapoint) remains valid for pattern comparison.

The low-redshift improvement per datapoint is $3.2\times$ larger than at high redshift. This matches the designed activation of $\Sigma(k, z)$ below $z = 0.5$ and provides a tomographic fingerprint consistent with a late-time lensing response.

4.5 Cosmological Parameter Shift

When Ω_m and A_s are varied on a 5×5 grid (not full MCMC), the results are shown in Table 4.

Table 4: Grid-optimized cosmological parameters. Both runs fix all nuisance parameters.

Model	Ω_m	$\ln A_s$	σ_8	S_8	$-\ln \mathcal{L}$
ΛCDM best	0.261	3.05	0.792	0.739	−346.26
Response ($\alpha = 0.10$)	0.261	2.90	0.734	0.685	−321.59

The response model prefers lower growth amplitude ($\sigma_8 = 0.734$ vs 0.792) because the enhanced lensing response provides the same shear signal with less underlying structure.

In Λ CDM coordinates, this *increases* the apparent tension with Planck ($S_8 = 0.685$ vs Planck ≈ 0.83). However, as discussed in Section 5, this tension is partly a coordinate artifact.

5 Reinterpreting S_8

Cosmic shear measures an effective amplitude proportional to the product of structure growth and lensing response:

$$A_{\text{WL}} \propto \sigma_8 \times \Sigma. \quad (4)$$

In Λ CDM, Σ is implicitly fixed to 1, so the entire shear amplitude is attributed to growth (σ_8/S_8). When Σ is allowed to differ from unity, the shear-relevant amplitude becomes

$$S_{\text{WL}} = \sigma_8 \Sigma(k_*, z_*). \quad (5)$$

Evaluated at the pivot $k_* = 0.15 \, h \, \text{Mpc}^{-1}$, $z_* = 0.25$:

Table 5: Shear amplitude decomposition.

Model	σ_8	Σ	S_{WL}	Planck S_8
Λ CDM	0.792	1.00	0.792	≈ 0.83
Response model	0.734	1.13	0.829	≈ 0.83

Note: S_{WL} evaluated at a single pivot point; a data-weighted effective Σ should be computed for robustness (see Section 6). The Planck value is itself Λ CDM-derived, not a direct shear measurement.

Although Λ CDM-style S_8 decreases from 0.739 to 0.685 (apparently worsening the Planck tension), the shear-relevant amplitude S_{WL} increases from 0.792 to 0.829. The latter is numerically close to the Λ CDM-derived Planck S_8 value, illustrating that part of the apparent discrepancy depends on how growth and response are parameterized. This demonstrates that the commonly discussed S_8 constraints depend on the assumption $\Sigma = 1$ and mix growth and response into a single parameter.

We stress that the Planck value $S_8 \approx 0.83$ is itself a Λ CDM-derived quantity, not an independent shear observation. The correct statement is: *in a decomposition that separates growth and lensing response, the same effective shear amplitude can arise from lower growth and higher response, so the apparent mismatch between CMB-inferred and shear-inferred amplitudes depends on the assumed parameterization.* A full resolution of whether this decomposition truly eliminates the tension requires consistent joint analysis with CMB data, which is beyond the scope of this work.

6 Limitations

This analysis has several important limitations:

Case A only. The response Σ modifies only the lensing amplitude, not the growth of structure (no μ modification). A fully self-consistent modified gravity implementation where Σ and μ are both present and coupled through the perturbation equations is

required to test whether the model can simultaneously improve the shear fit and affect growth in a physically consistent way.

No full posterior inference. All results are maximum-likelihood evaluations at fixed or grid-varied parameters. Full MCMC inference marginalizing over cosmological and nuisance parameters is needed to establish robust parameter constraints and proper statistical significance.

Single survey. Results are based on KiDS-1000 only. Cross-validation on independent datasets (DES Y3, KV450, HSC) is essential to confirm that the signal is not driven by survey-specific systematics.

Fixed nuisance parameters. Intrinsic alignment is set to zero and not marginalized. While this means the improvement cannot be IA compensation, it also means the interaction between Σ and IA has not been explored. A non-zero IA amplitude could absorb part of the Σ signal or enhance it.

Pivot-based S_{WL} . The effective shear amplitude is evaluated at a single pivot point ($k_* = 0.15 \, h \, \text{Mpc}^{-1}$, $z_* = 0.25$). A more robust definition using a data-weighted effective response,

$$\Sigma_{\text{eff}}^2 = \frac{\sum_{\ell} w_{\ell} C_{\ell}^{\Lambda\text{CDM}} \Sigma^2(\ell)}{\sum_{\ell} w_{\ell} C_{\ell}^{\Lambda\text{CDM}}}, \quad (6)$$

should be computed to verify that the decomposition is not sensitive to the pivot choice.

Parameterization dependence. The tanh form for $\Sigma(k, z)$ was chosen for mathematical convenience. Other functional forms with similar late-time, small-scale activation could yield different optimal parameters. The physical content is in the regime-activated structure, not the specific parameterization.

Illustrative Planck comparison. Because Planck S_8 is itself a ΛCDM -derived quantity, comparing S_{WL} to Planck S_8 is illustrative rather than a self-consistent joint analysis. A proper assessment requires joint inference across probes within a common theoretical framework.

7 Discussion

These results demonstrate that current shear data are sensitive to a separable lensing response degree of freedom. The improvement is robust to scale cuts, not reducible to amplitude rescaling, localized to low redshift, and consistent with a late-time activation. The smooth peak in the α scan and the tomographic fingerprint together provide strong evidence that the data genuinely prefer a scale- and redshift-dependent response rather than a simple rescaling.

The most important conceptual implication is that the S_8 constraints from CMB and weak lensing may be partly a *parameterization-dependent* comparison rather than a direct observational contradiction. In ΛCDM , the lensing amplitude is forced to map onto growth alone. Allowing an explicit response degree of freedom shifts the inferred growth amplitude while preserving the shear-relevant amplitude, indicating that the tension depends strongly on the assumption $\Sigma = 1$.

This does not mean the tension is “solved.” It means that the standard parameterization may hide a degree of freedom to which the data are sensitive. Whether this degree of freedom corresponds to real physics (modified gravity, screening mechanisms, entropy-gradient effects) or is absorbed by systematics in a full analysis remains to be determined.

The natural next steps are: (a) cross-survey replication on DES Y3 or KV450; (b) full MCMC inference with nuisance marginalization; (c) computation of a data-weighted Σ_{eff} ; (d) a Case B implementation where both Σ and μ are present; and (e) joint analysis with CMB and galaxy clustering data to test consistency across probes.

8 Conclusion

KiDS-1000 cosmic shear data favor a late-time, scale-dependent lensing response that improves the fit beyond Λ CDM while shifting inferred growth parameters. When growth and response are separated, much of the apparent S_8 tension is seen to arise from parameterization rather than direct observational contradiction. This suggests that future weak lensing analyses should consider models that allow independent growth and response degrees of freedom, and that the commonly used S_8 parameter may not be the most informative summary statistic when such degrees of freedom are present.

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